Frequency de-mixer: 'Unwanted Solo'

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Contents

1 Frequency De-Mixer **2**

2 Objective **2**

3 Background **2**

4 Problem Statement **2**

5 Method used: **3**

6 Tools and Techniques used **3**

6.1 Waveform 3
6.2 Spectrograms 4
6.3 Power Spectral Density (PSD) 5
6.4 Bode Plot 6
6.5 Pole Zero Plot 7

7 Results **7**

8 Conclusion **7**

1 Frequency De-Mixer

- Used for removing unwanted instrumental beats which are disrupting the intended harmony of a music track.

2 Objective

- To analyse the frequency characteristics of the input music audio.
- The aim is to recover the original song as faithfully as possible.
- Various tools are used in this process, such as:
  - Bode plots
  - Pole-zero plots
  - Other frequency analysis techniques

3 Background

- In music production, it is often necessary to restore and enhance audio recordings by removing undesirable elements.
- These unwanted elements could include:
  - Background noise
  - An instrument played out of sync
  - Any element that is not part of the desired audio
- Frequency demixing is a powerful technique that:
  - Uses frequency-domain analysis
  - Identifies and suppresses unwanted components in a mixed audio signal

4 Problem Statement

- An audio file is given that has been corrupted by the presence of an unwanted instrument, which is Piccolo, for the given input audio.
- We need to,
  - Analyse frequency characteristics of the input
  - Identify the frequency band that is occupied by the Unwanted instrument Piccolo and remove those frequencies while preserving other frequencies.
  - Export the cleaned version of the audio.

5 Method used:

Piccolo is an instrument which is played at high frequencies. So, my efforts were focused on removing those high-frequency components from the signal. At the same time, in order to ensure that other important audio elements remain intact, it was necessary to allow the frequencies corresponding to them to pass unchanged through the filter.

For this purpose, I used a **Butterworth band-stop filter**. This type of filter attenuates frequencies within a specific band while passing all others unaffected. The Butterworth filter is particularly useful in such scenarios due to its **maximally flat frequency response** in the passband, meaning it does not introduce any ripples or peaks. The **order of the filter** (i.e., the number of poles) determines how sharply the transition occurs between the passband and the stopband.

I experimented with filters of various orders. A filter with too low an order failed to sufficiently suppress the noise, while a very high order caused ripples near the lower frequencies of the passband. After testing different configurations, I found that an **11th-order Butterworth filter** struck the best balance between suppression and smoothness.

Next, to determine the most effective stopband range for removing the piccolo component, I tested various frequency bands ranging from 500 Hz to 10,000 Hz. I observed that the frequency range of **1000 Hz to 9000 Hz** was most effective in removing the unwanted piccolo noise, indicating that the instrument's sound was primarily concentrated within this range.

Hence, using an `order = 11` and a `stopband range = 1000--9000 Hz`, I designed the Butterworth band-stop filter. At the end, I evaluated my filtering approach using various tools and techniques to ensure the result preserved the desired audio quality while effectively suppressing the interfering piccolo component.

6 Tools and Techniques used

6.1 Waveform

- It is a graphical representation of how sound's amplitude changes with time. I used this tool to compare the input audio and filtered audio signals and see the difference between them.

For input audio

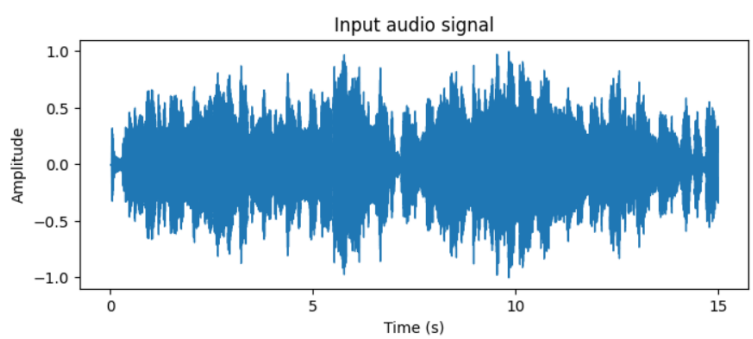

Figure 1: Input audio signal

For filtered audio

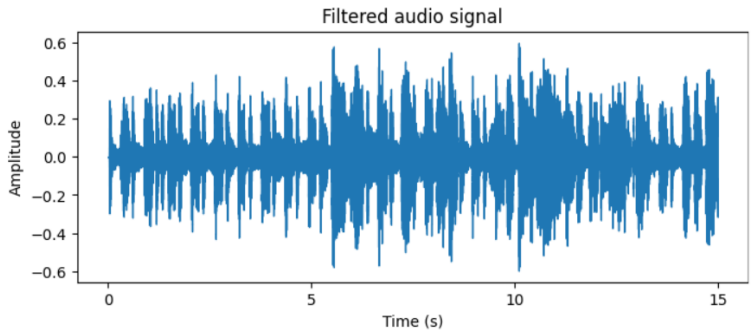

Figure 2: Filtered audio signal

We can see that some amplitudes at different times are suppressed in the filtered audio. Those correspond to the Piccolo instrument.

6.2 Spectrograms

- They are used to see how frequent The content of a signal changes over time. Again, these are used to compare the input audio and the output audio signal.

For input audio signal

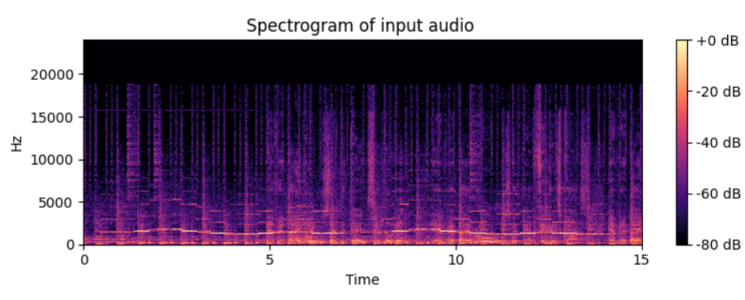

Figure 3: Spectrogram of input audio

For filtered audio signal

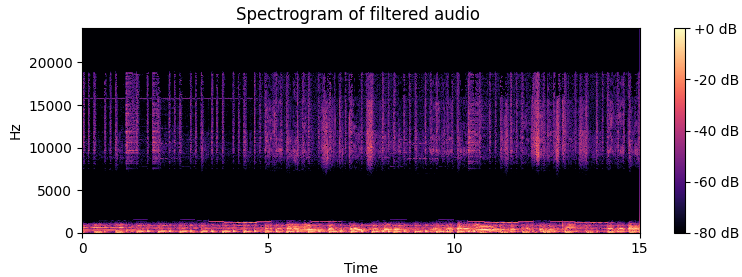

Figure 4: Spectrogram of filtered audio

Here we can see that frequencies between 1000-9000 are removed, which correspond to Piccolo instruments. Also, we can see that the rest of the frequencies are largely unchanged, as expected.

6.3 Power Spectral Density (PSD)

- It describes how the energy of a signal is distributed across different frequencies.

Let's see the input signal first:

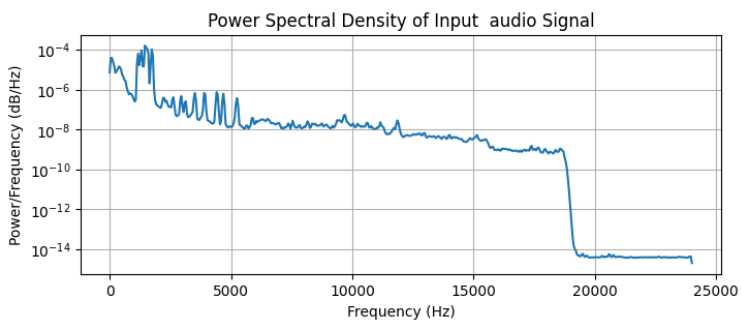

Figure 5: Power Spectral Density of Input Audio Signal

Here we can see that the energy of frequencies 1000 to 9000 is high, which is the reason why piccolo noise is too loud in the given input audio file.

Now let's see filtered audio's spectral density

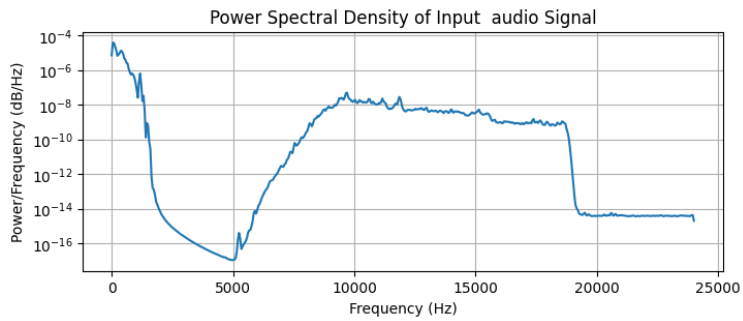

Figure 6: Power Spectral Density of Filtered Audio Signal

Reduction in power density magnitude between 0 to 10000 Hz suggests that the filter works correctly.

6.4 Bode Plot

- After applying the filter, the Bode plot is used to analyse the filter frequency response and ensure minimal distortion to passbands.

Let's see Bode plot for N(order) = 11 and Stopband range = 1000 Hz - 9000 Hz.

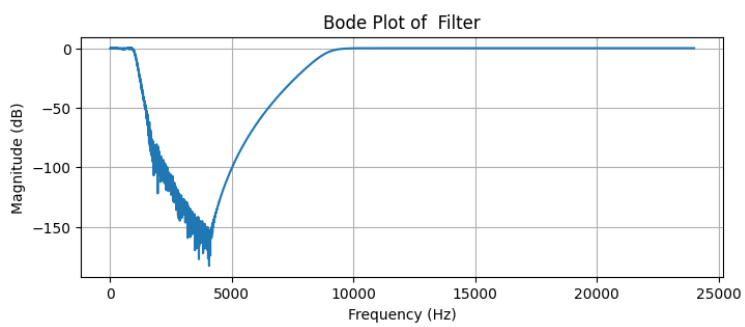

Figure 7: Bode Plot of Filter

- Here we can see that gain for frequencies near 0 and Greater than approximately 9000 is 0 dB, which means These frequencies are not changed by the filter.
- Whereas frequencies in between 1000 - 9000 Hz are suppressed as much as possible.
- I was unsure about whether the frequencies in the passband from 0 - 1000 Hz are actually unchanged or not, so to check it, I applied a function that gave me result as,
- **Passband (0–1000 Hz):**
  - **Min gain = 0.6284322329308625**
  - **Max gain = 1.0567897967966895**
  - **Mean gain = 0.9546582685730932**

- The gain is approximately close to 1, suggesting that they are hugely unchanged.

6.5 Pole Zero Plot

- It is majorly used to check stability of filter

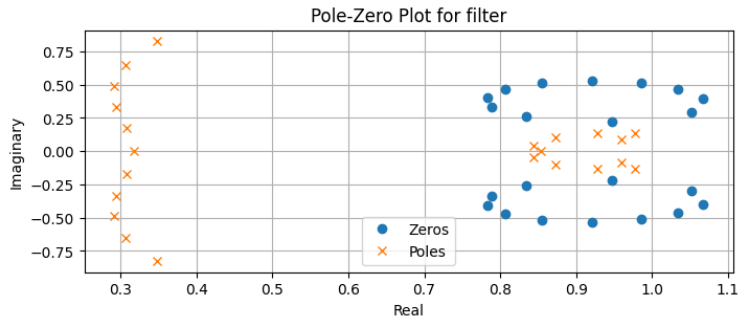

Figure 8: Power Spectral Density of Input Audio Signal

Here, as all poles lie within the unit circle, the filter is stable. Also it verifies that no ripples are created in the passband region.

7 Results

- Piccolo noise is significantly reduced while preserving other elements of audio. Cleaned audio is exported successfully.

8 Conclusion

- By applying appropriate filters and various tools, the unwanted instrumental beats were suppressed effectively. This shows a practical use of frequency demixing in music restoration.